

INSTALLATION INSTRUCTIONS

FOR REPLACEMENT COILS USED IN AIR HANDLERS

(-)CH: featuring Industry-Standard R-410A  Refrigerant



RECOGNIZE THIS SYMBOL AS AN INDICATION OF IMPORTANT SAFETY INFORMATION!

⚠ WARNING

These instructions are intended as an aid to qualified licensed service personnel for proper installation, adjustment and operation of this unit. Read these instructions thoroughly before attempting installation or operation. Failure to follow these instructions may result in improper installation, adjustment, service or maintenance possibly resulting in fire, electrical shock, property damage, personal injury or death.

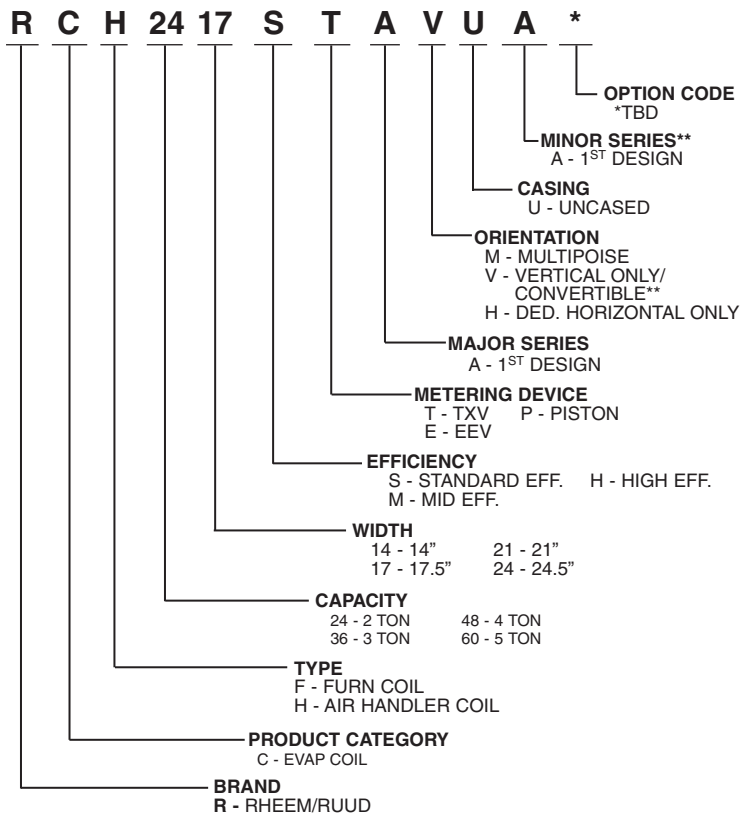


ISO 9001:2008

Certificate Number: 30164

DO NOT DESTROY THIS MANUAL

PLEASE READ CAREFULLY AND KEEP IN A SAFE PLACE FOR FUTURE REFERENCE BY A SERVICEMAN

FIGURE 1**MODEL NUMBER EXPLANATION**

"All replacement coils for HM and AU air handler applications will be HU."

**Convertible to horizontal using parts from original coil or using RXHH horizontal adapter kit.

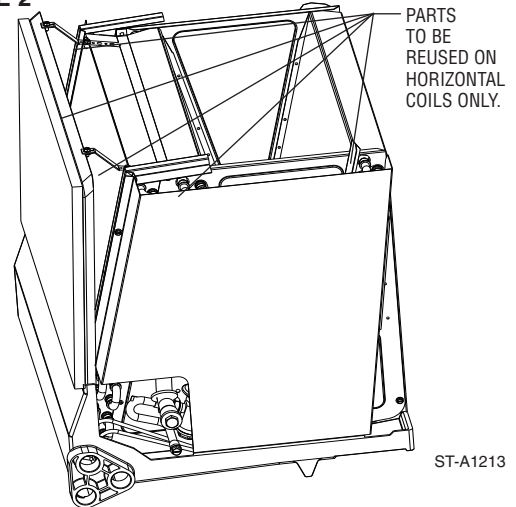
GENERAL

(-)CH is a replacement only uncased coil. (-)CH replacement coils are used only in (-)HKL, (-)HLL, (-)HLP, (-)HML, (-)HPL, (-)HPN, (-)HSL, (-)H1P, (-)H1T, (-)H2T, (-)H1V, and (-)H2V R410A air handlers.

For horizontal applications, a horizontal drip shield and water management parts (see Figure 2) must be installed to catch excess condensate drainage.

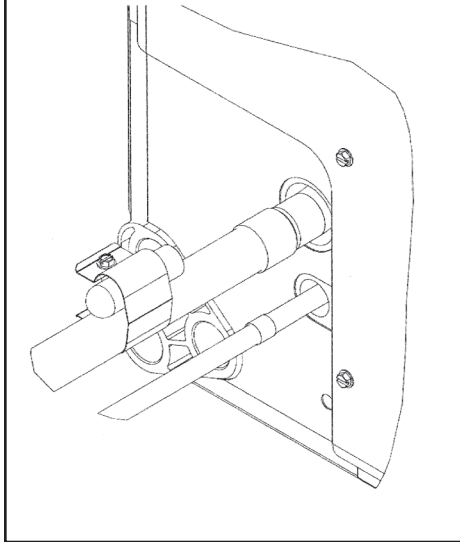
COIL END SHIELDS

All uncased replacement coils come equipped from the factory with sheet metal shields at the front and rear of the coil. The purpose of these shields is to isolate the aluminum tubing from copper residue left on the foil insulation by the original copper tube coil. Copper residue or copper oxide in

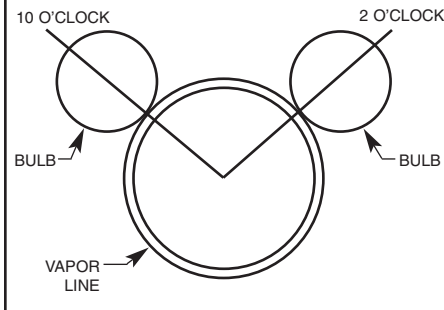
FIGURE 2**TABLE 1****R-410A UNCASSED COILS: (-)CSL- ACCESSORY INFORMATION**

Replacement R-410A Uncased Coil	Original Coil	Horizontal Adapter Kit Model Number
RCH2417STAVUA	RCSL-HU2417CU	RXHH-A02
RCH2421MTAVUA	RCSL-HU2621CU	RXHH-A03
RCH3617STAVUA	RCSL-HU3617CU	RXHH-A03
RCH3621STAVUA	RCSL-HU3621CU	RXHH-A03
RCH3621MTAVUA	RCSL-HU3821CU	RXHH-A04
RCH3624MTAVUA	RCSL-HU3824CU	RXHH-A04
RCH4821STAVUA	RCSL-HU4821CU	RXHH-A04
RCH4821MTAVUA	—	RXHH-A04
RCH4824STAVUA	RCSL-HU4824CU	RXHH-A04
RCH6024STAVUA	RCSL-HU6024CU	RXHH-A05
RCH2421HTAVUA	RCSN-HU2421CU	RXHH-A03
RCH3624HTAVUA	RCSN-HU3624CU	RXHH-A05
RCH4824HTAVUA	RCSN-HU4824CU	RXHH-A05
RCH6021STAVUA	—	RXHH-A06
RCH6024HTAVUA	RCSN-HU6024CU	RXHH-A05
RCH2417SPAVUA	RCSP-HU2417CU	RXHH-A02
RCH3617SPAVUA	RCSP-HU3617CU	RXHH-A03
RCH4821SPAVUA	RCSP-HU4821CU	RXHH-A04
RCH2417SPBMUA	—	RXHH-A02
RCH3617SPBMUA	—	RXHH-A03
RCH4821SPBMUA	—	RXHH-A04
RCH2417SEAVUA	—	RXHH-A02
RCH2421MEAVUA	—	RXHH-A03
RCH3617SEAVUA	—	RXHH-A03
RCH3621SEAVUA	—	RXHH-A03
RCH3621MEAVUA	—	RXHH-A04
RCH6021SEAVUA	—	RXHH-A06
RCH6024MEAVUA	—	RXHH-A05

**FIGURE 3
BULB LOCATION**



**FIGURE 4
BULB LOCATION**



contact with the aluminum tubing in the presence of moisture will result in galvanic corrosion and leaks in the aluminum tube at the contact point. The shields must be in place on the coil when replacing a copper tube coil to prevent the galvanic corrosion.

REFRIGERANT CONNECTIONS

Keep the coil connections sealed until refrigerant connections are to be made. See the Installation Instructions for the outdoor unit for details on line sizing, tubing installation, and charging information.

Coil is shipped with a low (5 - 10 PSIG) pressure charge of dry nitrogen. Evacuate the system before charging with refrigerant.

Install refrigerant tubing so that it does not block service access to the front of the unit.

Nitrogen should flow through the refrigerant lines while brazing.

Use a brazing shield to protect the cabinet's paint from being damaged by torch flames.

After the refrigerant connections are made, seal the gap around the connections with pressure sensitive gasket. If necessary, cut the gasket into two pieces for a better seal.

TEV SENSING BULB

IMPORTANT: DO NOT perform any soldering with the TEV bulb attached to any line.

After soldering operations have been completed, clamp the TEV bulb securely on the suction line at the 10 to 2 o'clock position with the strap provided in the parts bag.

Insulate the TEV sensing bulb and suction line with the provided pressure sensitive insulation (size 4" x 7") and secure with provided wire ties.

IMPORTANT: TEV sensing bulb should be located on a horizontal section of copper suction line, just outside of coil box. The copper sensing bulb must never be placed on any aluminum tube as this will result in galvanic corrosion and eventual failure of the aluminum tube.

ELECTRONIC EXPANSION

The RCF EXV equipped coils, cased and uncased, are the first Rheem indoor products to be produced with the noncommunicating, stand-alone EXV control. One of the biggest advantages of an EXV is the control can intelligently change the EXV position based on system demands other than just suction line temperature. By the measurement of the suction pressure via the vapor line pressure transducer (factory installed) and the vapor line thermistor (field connected to the vapor line, but factory provided within the air handler) the EcoNet™ enabled air handler control calculates the suction superheat at the indoor coil. This calculation permits the air handler control to make decisions for when to open and close the electronic expansion valve for the purpose of maintaining a predetermined suction superheat. The electronic valve is equipped with a 4-pole removable external stator, and inlet and outlet Chatleff fittings for optimal serviceability. These valves also have an internal check valve to provide heat pump compatibility. When operating in heating mode, the air handler control will open the electronic valve completely to permit the check valve to operate and maximize reverse refrigerant flow.

EXV VAPOR LINE THERMISTOR

⚠ NOTICE

THE VAPOR LINE THERMISTOR DOES NOT COME WITH THE REPLACEMENT COILS. REMOVE THE THERMISTOR FROM THE ORIGINAL COIL AND INSTALL ON THE NEW COIL.

IMPORTANT: DO NOT perform any brazing with the vapor line thermistor attached to any line. After brazing operations have been completed, clamp the vapor line thermistor securely on the vapor line at the 10 to 2 o'clock position with the clip provided on the thermistor. Insulate the vapor line thermistor and vapor line with the provided pressure sensitive insulation (size 4" x 7") and secure with provided wire ties.

Make sure to protect the EXV pressure transducer, vapor thermistor, copper to aluminum joint, and service valves from overheating by use of wet rag or some type of shielding. Double tip torches are not recommended.

IMPORTANT: Vapor line thermistor should be located on a horizontal section of vapor line, just outside of coil box. The copper thermistor must never be placed on any aluminum tube as this will result in galvanic corrosion and eventual failure of the aluminum tube.

IMPORTANT: Never place the thermistor on the heat effected zone near the braze connection, but it should be located within 6" of the indoor unit.

FIGURE 5
THERMISTOR LOCATION

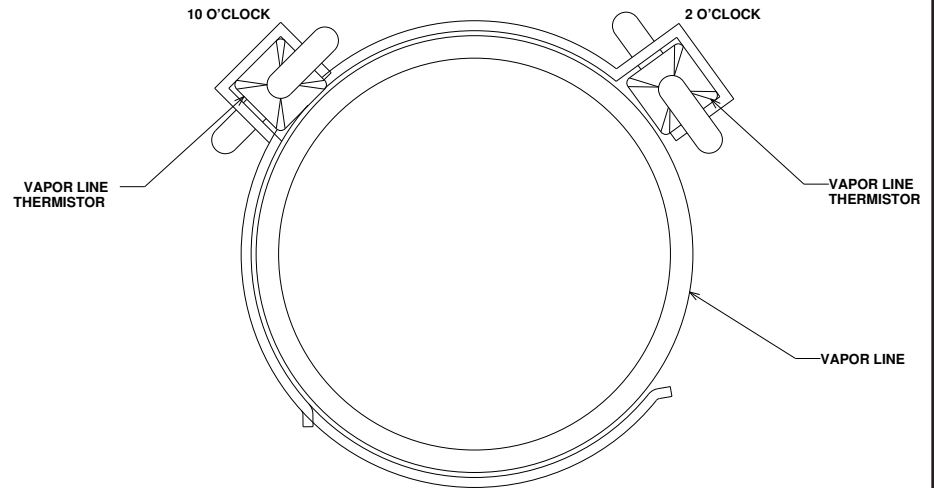
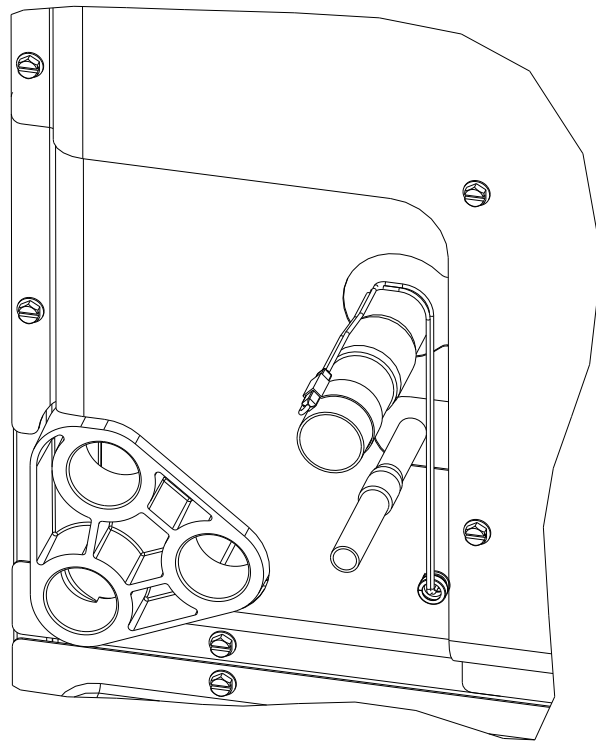


FIGURE 6
THERMISTOR LOCATION



FACTORY PROGRAMMED SUPERHEAT

The stand alone EXV control is pre-programmed for 10°F superheat. The following dip switch settings must be set at the time of coil installation.

Air Handler	Outdoor Unit	Superheat (°F)	Dipswitch Settings	
			1	2
RCF2417SE	RP1724	6	ON	OFF
RCF2421ME	RA1724, RA2024, RP2024	6	ON	OFF
RCF3617SE	RP1736	6	ON	OFF
RCF3621ME	RA1736	8	OFF	ON
RCF6021SE	RA1748 RA2036, RA2048, RA2060 RP2036, RP2048, RP2060	6	ON	OFF
RCF6024ME	RA1660	6	ON	OFF

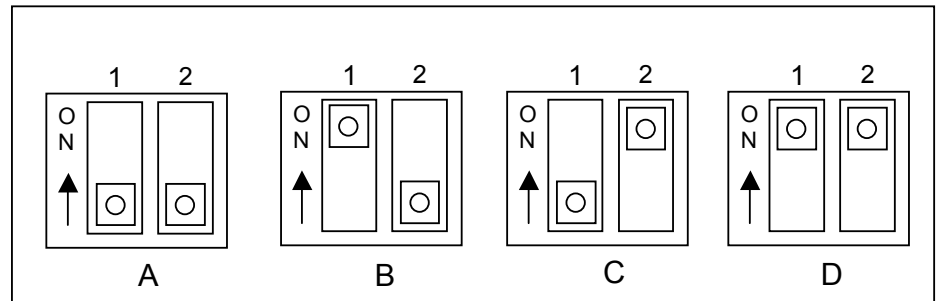
SUPERHEAT OFFSET DIP SWITCH SETTINGS

Although the above superheat set point is considered to be the most efficient set point for each air handler, installation conditions can drastically effect the measurement of superheat by the air handler control. For this reason the following dip switch settings have been provided to enable flexibility for various installation conditions.

Superheat Offset Selection Profile	Superheat Setting (°F)
A	10
B	6
C	8
D	12

EXV STEP DIP SWITCH

The EXV dip switch has an optional 500 or 1600 steps setting. This dipswitch should currently be in the 500 step position only. Rheem does not currently supply 1600 step EXV's. The dipswitch makes the control forward compatible with a 1600 step EXV for possible future use.



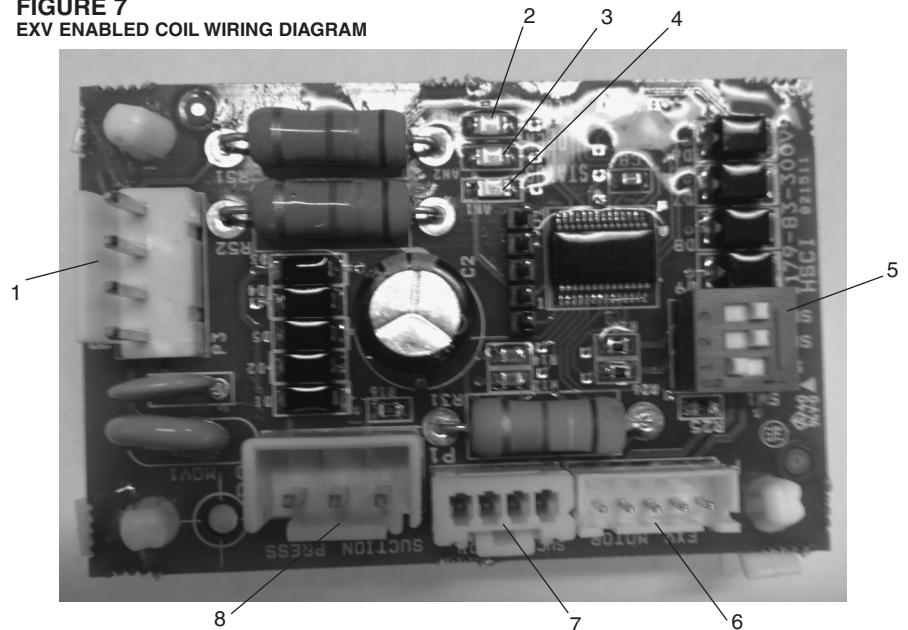
DIAGNOSTICS

There are 2 LEDs (green/yellow) on the standalone EXV control which indicate valve movement. When the green LED is illuminated, the control is moving the valve in the open direction. When the yellow LED is illuminated, the control is moving the valve in the closed direction. When neither LED is illuminated, the valve is not being moved by the control. In addition to the diagnostic lights on the control, it is possible to feel the coil on the EXV pulse when the control is attempting to change the EXV position.

Status LED	Board Fault
1	Only suction temperature valid – suction pressure nor coil temperature are valid
2	No Valid Suction Temperature
3	Valve near open position
4	Suction pressure out of range

“EXV STANDALONE CONTROL” PHYSICAL INTERFACE

FIGURE 7
EXV ENABLED COIL WIRING DIAGRAM



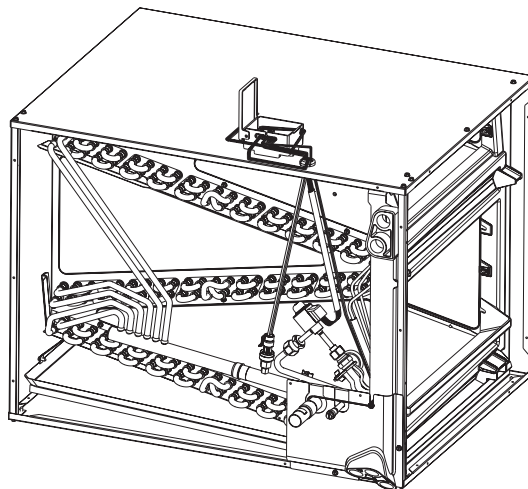
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|---|---|
| 1. Thermostat Connection | 6. Electronic Expansion Valve Connection |
| 2. Yellow LED (close) | 7. Suction and Coil Thermistor Connection |
| 3. Green LED (open) | 8. Suction Pressure Transducer Connection |
| 4. Red LED (status) | |
| 5. Dip Switch (EXV steps, 2 & 3 SH adjustment dip switch) | |

EXV STANDALONE CONTROL OPTIONAL MOUNTING LOCATIONS

NOTICE

THE STANDALONE EXV CONTROL, VAPOR LINE THERMISTOR, PRESSURE TRANSDUCER, AND WIRING HARNESSES DO NOT COME WITH THE UNCASSED REPLACEMENT COILS. REMOVE THESE COMPONENTS FROM THE ORIGINAL COIL AND INSTALL ON THE NEW COIL.

FIGURE 8
OPTIONAL CONTROL INSTALLATION LOCATION

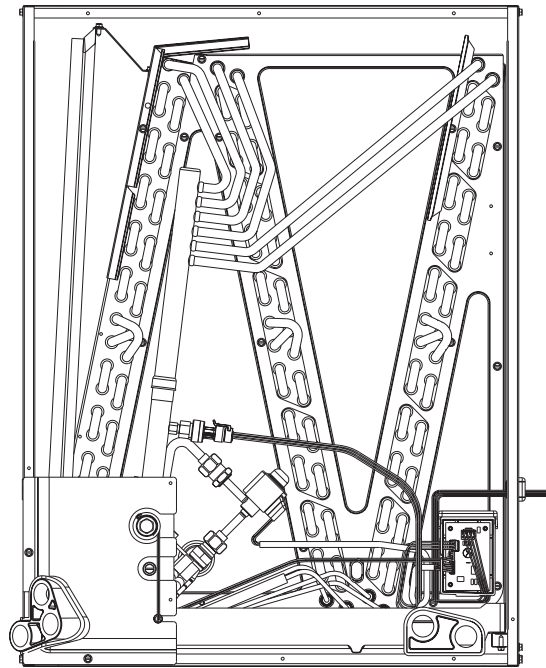


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The EXV standalone control and housing is factory installed inside the case on the tube sheet. This location has been tested and approved for long term operation inside the highly humid environment. In the event servicing the EXV control without opening the coil door or to see the operational lights while the system is in operation the following alternate control and housing location is approved. The mounting location will require the wiring harnesses to be disconnected from the control, the control removed from the housing and the housing mounting screws removed. The wires will need to be routed through the knockout (there is one knockout per side) and then the assembly reassembled externally to the coil case.

IMPORTANT: It is recommended to place aluminum tape over the screw holes in the tube sheet to prevent bypass air.

FIGURE 9
FACTORY CONTROL INSTALLATION LOCATION



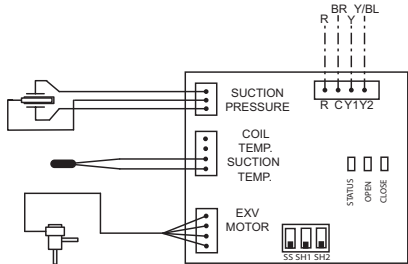
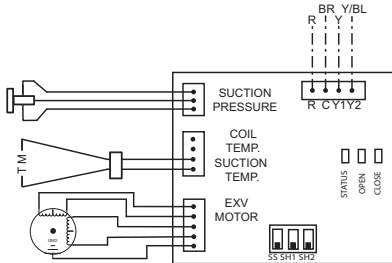
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EXV STANDALONE WIRING

⚠ NOTICE

THE STANDALONE EXV CONTROL, VAPOR LINE THERMISTOR, PRESSURE TRANSDUCER, AND WIRING HARNESSES DO NOT COME WITH THE UNCASSED REPLACEMENT COILS. REMOVE THESE COMPONENTS FROM THE ORIGINAL COIL AND INSTALL ON THE NEW COIL.

For proper operation the standalone EXV control requires 24VAC power and staged operation thermostat signals. The following diagram should be used to connect the flying leads provided with the cased coil to the thermostat wiring. When employed with the Econet™ Communicating System attach the flying leads provided with the cased coil to the legacy thermostat outputs on the Econet™ compatible IFC.

WIRING DIAGRAM 	NOTES 1..... CONNECTORS SUITABLE FOR USE WITH COPPER CONDUCTORS ONLY. 2..... LOW VOLTAGE CIRCUIT TO BE N.E.C. CLASS 2 WITH A CLASS 2 TRANSFORMER 24 VOLT, 50 OR 60 HZ. MIN 18 AWG. 3..... TO THERMOSTAT, REFER TO SYSTEM SCHEMATICS OR SCHEMATICS ON FURNACE FOR LOW VOLTAGE CONTROL WIRING. <td> WIRING INFORMATION LINE VOLTAGE -FACTORY STANDARD -FACTORY OPTION -FIELD INSTALLED LOW VOLTAGE -FACTORY STANDARD -FIELD INSTALLED REPLACEMENT WIRE -MUST BE THE SAME SIZE AND TYPE OF INSULATION AS ORIGINAL (105C. MIN.) WARNING -CABINET MUST BE PERMANENTLY GROUNDED AND CONFORM TO I.E.C., N.E.C., C.E.C., NATIONAL WIRING REGULATIONS, AND LOCAL CODES AS APPLICABLE. </td>	WIRING INFORMATION LINE VOLTAGE -FACTORY STANDARD -FACTORY OPTION -FIELD INSTALLED LOW VOLTAGE -FACTORY STANDARD -FIELD INSTALLED REPLACEMENT WIRE -MUST BE THE SAME SIZE AND TYPE OF INSULATION AS ORIGINAL (105C. MIN.) WARNING -CABINET MUST BE PERMANENTLY GROUNDED AND CONFORM TO I.E.C., N.E.C., C.E.C., NATIONAL WIRING REGULATIONS, AND LOCAL CODES AS APPLICABLE.
SCHEMATIC DIAGRAM 	COMPONENT CODES CLOSE.....YELLOW LED COIL TEMP.....COIL INLET THERMISTOR CONNECTION COMP.....COMPRESSOR EXV (5 PIN).....ELEC. TX VALVE ODF.....OUTDOOR FAN CONNECTION OPEN.....GREEN LED SH1.....SUPERHEAT OFFSET SWITCH 1 SH2.....SUPERHEAT OFFSET SWITCH 2 SS.....EXV STEP SELECTION (MUST REMAIN OFF) STATUS.....RED LED SUCTION PRESSURE.....SUCTION PRESS. TRANSDUCER CONN. SUCTION TEMP.VAPOR LINE THERMISTOR CONN. TM.....THERMISTOR RVS.....REVERSING VALVE CONN.	WIRE COLOR CODE BK.....BLACK G.....GREEN PR.....PURPLE BR.....BROWN G Y.....GRAY R.....RED BL.....BLUE O.....ORANGE W.....WHITE Y.....YELLOW ELECTRICAL WIRING DIAGRAM NON-COMMUNICATING STANDALONE EXV CONTROL W/ PRESSURE TRANSDUCER PART NO.: 90-101897-09 REV: 00

FLOW CHECK PISTON

NOTICE

FOR PROPER SYSTEM OPERATION, IT MAY BE NECESSARY TO REPLACE THE PISTON INSTALLED IN THE INDOOR COIL. CHECK THE SERVICE VALVES ON THIS UNIT TO SEE IF A NOTICE TAG ALONG WITH A PLASTIC BAG CONTAINING A PISTON IS ATTACHED. IF ONE IS PRESENT A CHANGE OF THE PISTON IS REQUIRED. FAILURE TO CHANGE THE PISTON CAN RESULT IN IMPROPER PERFORMANCE OF THE SYSTEM.

The flow check piston is a multi-purpose device. With flow into the compression nut end from the liquid line, the piston is in a check position and acts as the expansion device with flow through the metering orifice in the center of the piston. The “O” ring on the end of the piston prevents refrigerant from bypassing the metering orifice. Flow from the metering orifice is centered into a distributor which serves to evenly distribute refrigerant to the evaporator circuits. With flow in the reverse direction (direction of arrows on the distributor body), the piston is forced off the seat and liquid from the condenser is allowed to free flow around the piston.

It is essential that the heat pump indoor and outdoor sections be properly matched. Use only matched components as shown in sales specification sheets.

A piston size that is too small will cause starving and one that is too large will cause flooding. In either case, system performance, reliability and charge balance (heating and cooling) will be unacceptable.

Change the piston in the distributor on the indoor coil before installing the coil and charging the system following the procedure below:

- Using a back-up wrench on the distributor body, loosen the compression nut to gain access to the piston.
- Using the wire provided with replacement pistons, run (hooked end) through hole in piston.
- Hook nose end of piston and lift gently from distributor body.
- Replace piston with one of proper size (see Table 4), install piston with gasket end of piston in distributor. Do not force piston into distributor.
- **NOTE:** With piston in distributor, seal end should be down and should not be seen looking in end of distributor. Piston must be free to rotate and move up and down. Make sure piston is free to move in distributor body.
- Insure distributor gasket is located properly in the distributor body.

- Replace compression nut using back-up wrench on distributor body. Torque compression nut end with 8 to 10 ft. lbs.
- Original piston size is stamped on outside of distributor body. Remove new piston size label from poly bag new piston came in and install new size label on outside of distributor tube.
- Check fittings for leaks after installation, evacuation and charging is complete.

IMPORTANT: Do not attempt to drill pistons to size in the field. Metering holes have a special chamfered inlet and cannot be modified.

IMPORTANT: Do not replace the neoprene "O" ring on the piston with any type of seal. Contact the parts department for the exact replacement of "O" ring.

FIGURE 10
PISTON AND DISTRIBUTOR ASSEMBLY

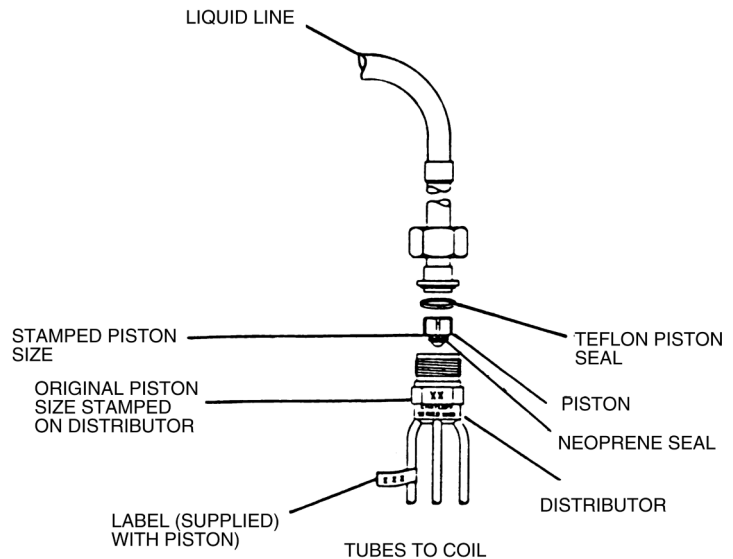


TABLE 2
R-22 TXV CONVERSION KITS (TO BE USED TO CONVERT R-410A COIL TO OPERATE WITH R-22)

AIR HANDLER CROSS REFERENCE CHART			
ORIGINAL AIR HANDLER	ORIGINAL AIR HANDLER COIL	RECOMMENDED ALUMINUM TUBE REPLACEMENT COIL (R-410A)	R-22 TXV CONVERSION KIT MODEL NO.
RHSA-**1817	RCSA-**2417	RCH2417STA	RXCT-HBA
RHSA-**2417	RCSA-**2417	RCH2417STA	RXCT-HBA
RHLA-**2417	RCSA-**2417	RCH2417STA	RXCT-HBA
RHSA-**3017	RCSA-**3617	RCH3617STA	RXCT-HBB
RHSA-**3617	RCSA-**3617	RCH3617STA	RXCT-HBB
RHLA-**3617	RCSA-**3617	RCH3617STA	RXCT-HBB
RHSA-**3621	RCSA-**3621	RCH3621STA, RCH2421MTA, or RCH2421HTA	RXCT-HBB
RHSA-**4221	RCSA-**4821	RCH4821STA or RCH3621MTA	RXCT-HBC
RHSA-**4821	RCSA-**4821	RCH4821STA or RCH3621MTA	RXCT-HBC
RHLA-**4821	RCSA-**4821	RCH4821STA or RCH3621MTA	RXCT-HBC
RHSA-**4824	RCSA-**4824	RCH4824STA or RCH4824MTA	RXCT-HBC
RHLA-**4824	RCSA-**4824	RCH4824STA or RCH4824MTA	RXCT-HBC
RHSA-**6024	RCSA-**6024	RCH6024STA, RCH6024HTA, RCH4824HTA, or RCH3624HTA	RXCT-HBD
RHLA-**6024	RCSA-**6024	RCH6024STA, RCH6024HTA, RCH4824HTA, or RCH3624HTA	RXCT-HBD

**= AU, HM, OR HU

